

Computer Organization and Architecture · 21CSS201T

Unit 1 · Session 1 · SLO 2 — Solved Worksheet

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Unit – 1 : Session – 1 : SLO – 2 | Types of computers

■ Multiple Choice Questions

1. c — Supercomputer
2. c — High throughput and multi-user support
3. b — Built into another device for a specific task
4. c — Personal laptop
5. c — Speed and reliability

■ Two-Mark Questions

1. A minicomputer is a mid-range, multi-user computer smaller than a mainframe but more powerful than a microcomputer; it supports several terminals at once. Real-time application: process control in a factory / college server.
2. General-purpose vs special-purpose: (i) General-purpose computers run many different applications (PCs); special-purpose computers are designed for one dedicated task. (ii) General-purpose machines are flexible but less optimized; special-purpose machines are faster/cheaper for their single task but not reprogrammable for other work.

■ Scenario-Based Long Answer

- Recommended: a Mainframe-class server (a high-end minicomputer is the budget alternative).
- Justification: The university needs simultaneous access by many users (student records, exam processing, library). Mainframes are built exactly for this: very high throughput, strong multi-user support, large memory and I/O capacity, high reliability/availability, and transaction processing at scale. A minicomputer could serve a smaller load, but for a mid-sized university with concurrent access across departments, a mainframe's multi-user capability and reliability make it the most suitable choice.

■ Match the Following

- Supercomputer — c) IMD weather forecasting
- Mainframe — large multi-user transaction servers (e.g., bank/airline reservation systems)
- Minicomputer — b) College server
- (Smartwatches — a) — are examples of embedded/microcomputer systems.)

■ Fill in the Blanks

1. Supercomputer
2. Special-purpose
3. Embedded
4. microcomputers
5. Ubiquitous (pervasive)

■ Scenario-Based Small Activity

- 5 devices with embedded computers: smartphone, washing machine, microwave oven, smart TV, car (ECU).
- Example — washing machine: its embedded controller runs a fixed control program; it reads sensors (water level, load, temperature), times each wash/rinse/spin stage, and drives the motor and valves automatically. It

is special-purpose: it does only washing control, reliably and cheaply.

■ **True or False**

1. FALSE — Microcomputers serve one/few users.
2. FALSE — Special-purpose are less flexible.
3. TRUE
4. FALSE — Minicomputers serve departments/organizations, not national defense scale.
5. TRUE